# KHALIDUZZAMAN MREDUL

Phone: +8801971170798 || Email: Khaliduzzaman.mredul@gmail.com || Address: Bashundhara R/A, Dhaka - 1229, Bangladesh

## Personal Summary

AI enthusiast with 3+ years of experience building scalable AI/ML systems in the Computer Vision and NLP domains. Specialized in transforming advanced research into industry-standard embedded products and IoT solutions. Possess a deep interest in robotics and intelligent systems with noteworthy competitive achievements.

## PROFESSIONAL EXPERIENCE

### D2A2I Institute of Innovation, American International University-Bangladesh (AIUB)

***AI/ML Research Engineer [Computer Vision AI]***

*Sep. 2023 - Present*

- Led the R&D team of nation-wide deployed **"Real-time Object Detection based TRP Measurement System"** project for **Bangladesh Satellite Company Limited (BSCL).**
- Developed the second iteration of the visual detection algorithm using **State of the art (SOTA)** object detection models that achieved **15% better accuracy with nearly 65% smaller dataset**, increasing both dataset preparation and training efficiency compared to the existing one.
- Created a fully custom annotated dataset of more than **250K image samples across 264 classes** of local and international TV Channel Logos for training and performance evaluation of multiple SOTA models for specific use case scenario.
- Working on the development of a **Smart-sensor Attachment Module** designed for safety enhancement of power wheelchair users, based on **Realtime Object Detection and Laser Sensing (LIDAR, ToF, RGB-D)** for Canadian native environment.

***Research Assistant [AI | Robotics]***

*Feb. 2023 - Aug. 2023*

- Contributed to the design and development of a **Half-body Humanoid Desk Assistant Robot** featuring **LLM and Computer Vision** for voice feedback and optical visualization for basic university guidance and help of newcomers
- Conducted **performance evaluation of different deep learning models** and analyzed feasibility for intended use case
- Supervised and managed production grade **Linux operated Single Board Computer (Raspberry Pi, Jetson Orin Nano)** driven hardware devices to ensure optimal functionality for AI applications
- Contributed to manual image annotation for training and validation of large-scale datasets for production level deployment.

### Sonzaikan LTD, London, UK

***Full Stack AI Developer & Product Designer [Remote]***

*Mar. 2025 – Present*

- Designed and developed an **AI-powered smart typewriter device named "Tagore"** (https://writetagore.com) intended to distraction free rigorous writing experience for individuals such as scientific writers, authors, journalists and students.
- Developed an LLM backend "Author Intelligence" for basic writing operations like paraphrasing, concept mapping, tone changing etc.
- Collaborated with **University of Oxford and University of Sheffield** to fabricate the initial prototype of the device.
- Implemented a custom embedded Linux system for the device to be used as a proprietary product.

## SKILLS

**Languages:** Python, C/C++, Java Script, SQL
**Frameworks:** PyTorch, Tensorflow, FastAPI, Django, React
**Tools/OS:** Docker, GitHub, Tauri, Buildroot, Yocto, ROS2, Linux
**Database**: MySQL, PostgreSQL, SQLite
**Design & Simulation**: Autodesk Fusion 360, Blender, Gazebo, Webots, COMSOL

## ACADEMIC HISTORY

### American International University-Bangladesh (AIUB)

*Bachelor of Science in **Computer Science Engineering***

- ***Major:*** *Computer Engineering*
- ***Graduation Year:*** *2023*

***Award:*** *Chairman's Award (GOLD)*

## EXTRACURRICULAR ACTIVITIES

### AIUB Robotic Crew (ARC)

- ***Team Captain***
  *Sep. 2022 – Aug. 2023*
- ***Team Lead (Software)***
  *Sep. 2021 – Aug. 2022*

## PUBLICATIONS & PERSONAL PROJECTS

**Publication: "**IR-Based Automation System with Telegram Bot Integration", ieeexplore.ieee.org/document/10914456
**Undergoing Research:** Design & development of Finless Maneuver System for Projectiles at Subsonic to Supersonic velocity by manipulating the intake.
**Completed Projects:**

- Autonomous Visual & GPS Navigation System for Mars Rover
- Voice Controlled Power wheelchair for people with physical disability
- Vision based Abnormal Anomaly Detection System inside ATM Booth
- Autonomous Solar Panel Cleaning System based on visual dust detection

## CIRTIFICATION

**CISCO**

***CyberOps Associate***

https://www.credly.com/badges/fb228585-240f-4c7a-a945-a759572dc276?source=linked_in_profile
Cisco verifies the earner of this certificate successfully completed the CyberOps Associate course. The holder of this credential has a broad understanding of Security Operations. They developed problem-solving skills to detect and analyze intrusions and monitor the network.